Tabular Foundation Models for Data Science Agents

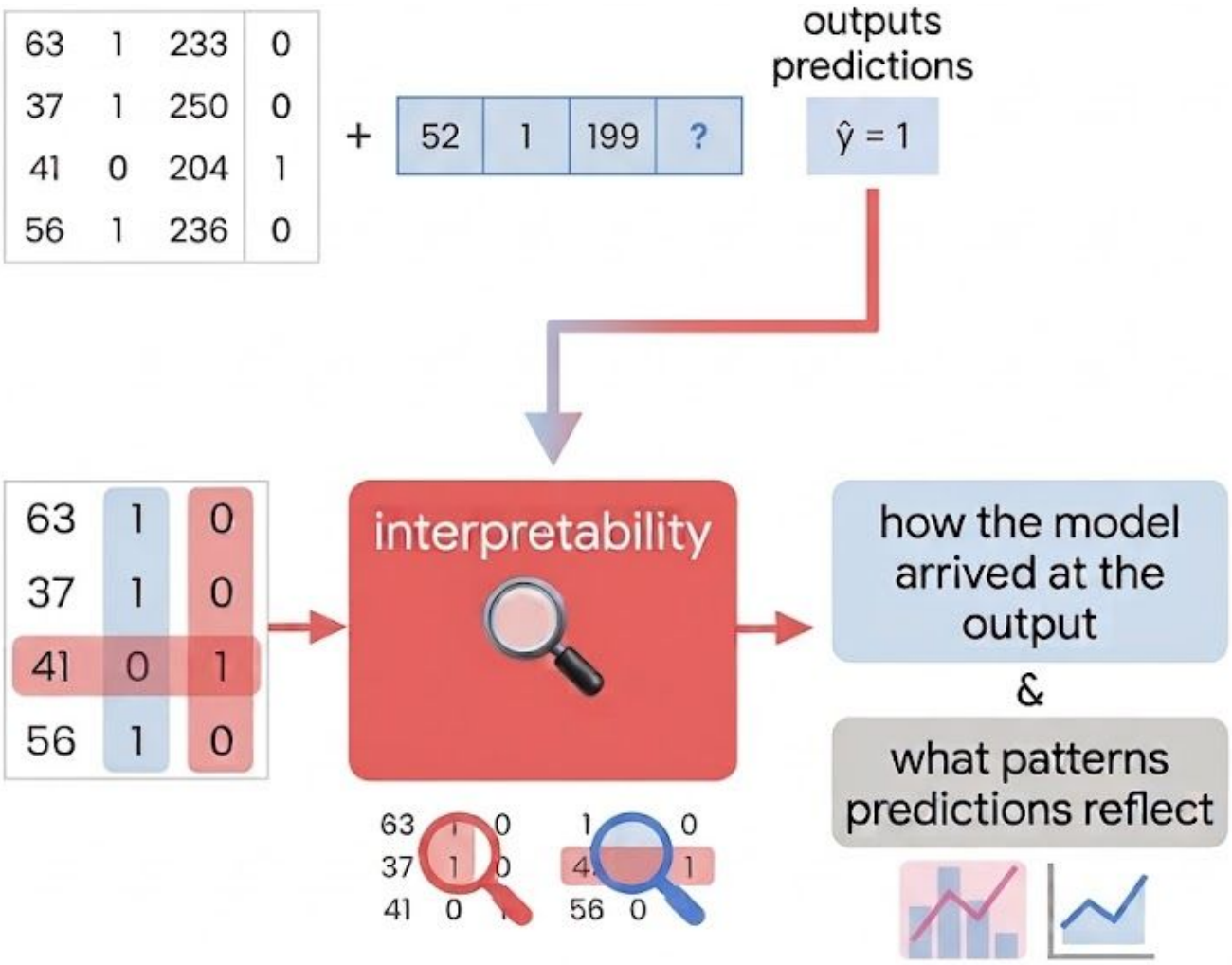
Interpreting TabPFN predictions

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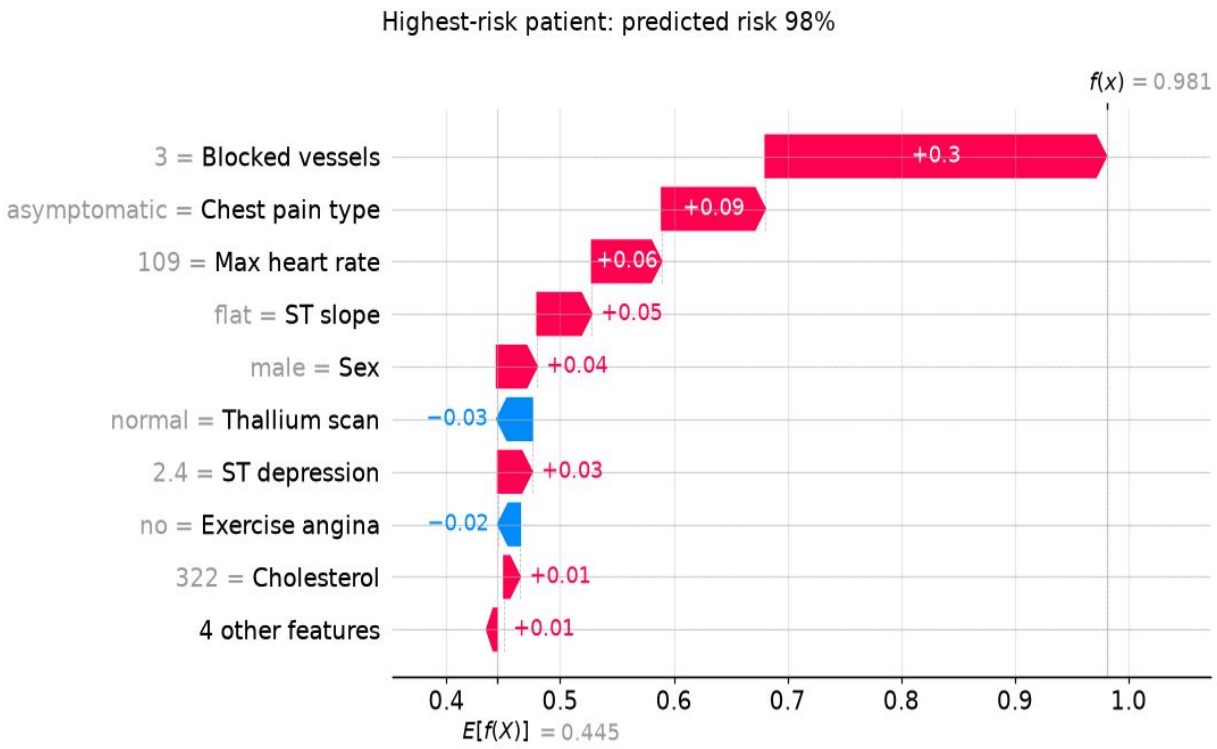


Interpreting TabPFN predictions



What feature values explain a single prediction?

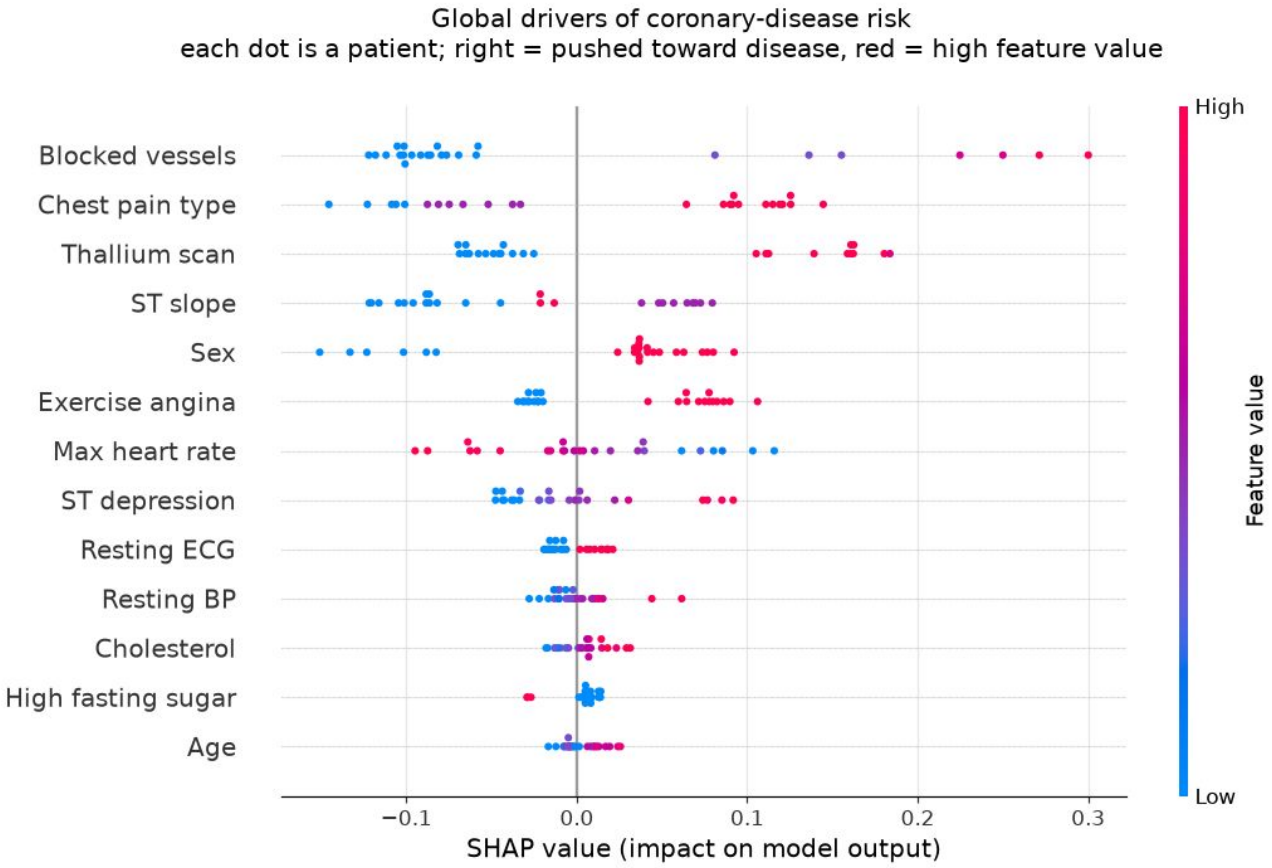
SHAP Value: estimated contribution of a feature value to a model prediction



Which features does the model use?

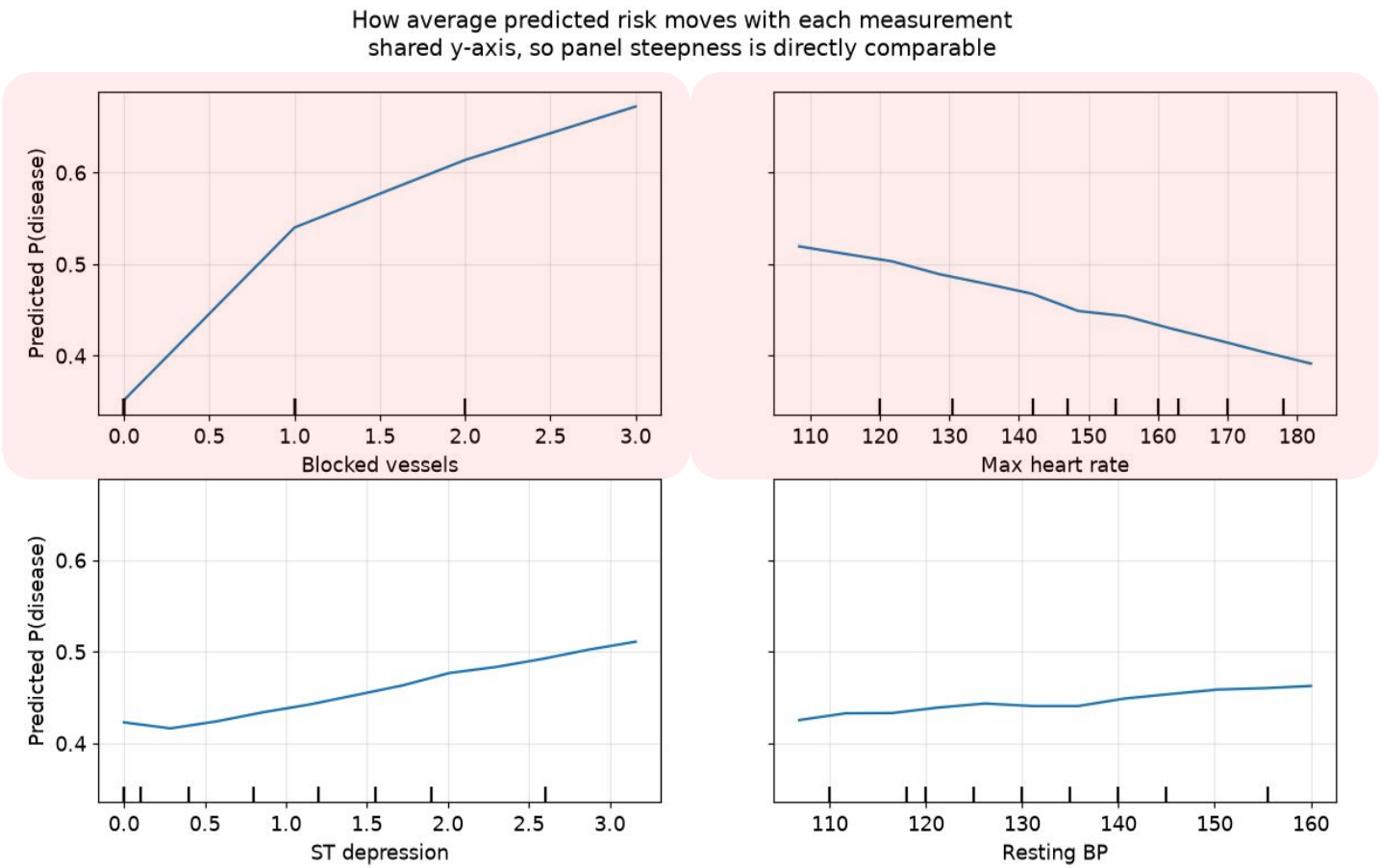
We can also plot all SHAP values for all features and records.

- This gives us a high-level summary of feature importance.



What is the impact of these features on model predictions?

Partial Dependence Plots: show the relationship between the predicted probability and continuous features.



Interpreting TabPFN predictions

Three methods to interpret TabPFN predictions:

- **Explain individual prediction** with shad values and waterfall plot
- **Plot all the shad values** on visual plot to get better idea of feature importance.
- **Partial dependence plots** that shows how the model responds to changes in individual feature values.

Tabular Foundation Models for Data Science Agents

Real-time inference with the KV cache

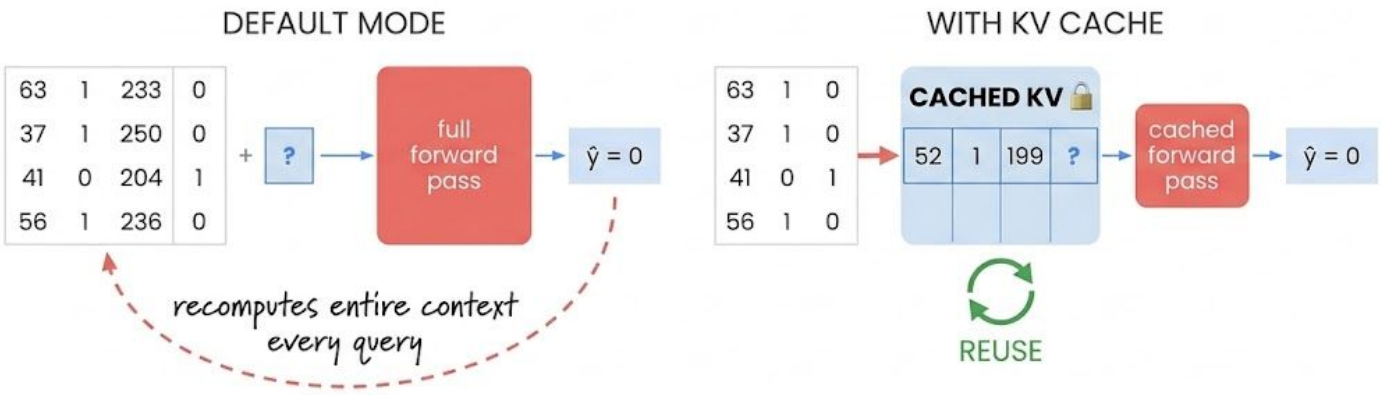
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Real-time inference with the KV cache

TabPFN Key Value (KV) Cache to reuse a fixed training context and measure its impact on request latency, the time it takes for TabPFN for output a prediction.



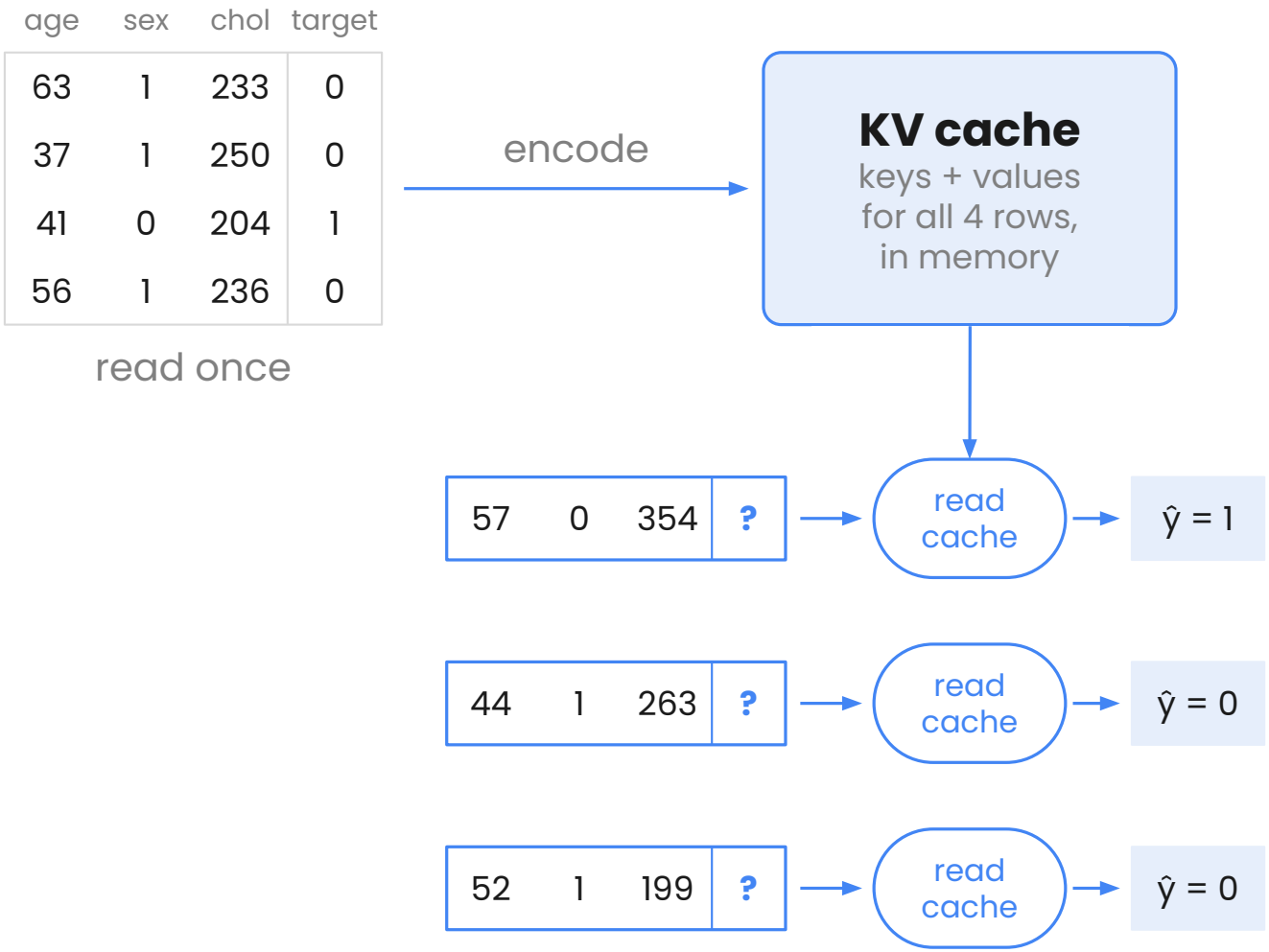
Inference without KV cache

	age	sex	chol	target
	63	1	233	0
	37	1	250	0
1	41	0	204	1
	56	1	236	0
			+	57 | 0 | 354 | ? → forward pass → ŷ = 1
	63	1	233	0
	37	1	250	0
2	41	0	204	1
	56	1	236	0
			+	44 | 1 | 263 | ? → forward pass → ŷ = 0
	63	1	233	0
	37	1	250	0
3	41	0	204	1
	56	1	236	0
			+	52 | 1 | 199 | ? → forward pass → ŷ = 0

3×
full forward passes

32 / 48
cells re-read for nothing

Inference with KV cache



1×

encode, then cheap reads

16 / 16

cells read, none wasted

When to use the KV-cache?

✓ Use it

- Real-time inference
- Low latency is critical
- Fixed or slowly changing training set

✕ Skip it

- Batch workflows in which predictions are done in one go
- Training set changes often or is used once

Rule of thumb: many predictions on the same training set? Cache it.